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20CS6T01 - COMPILER DESIGN				

COURSE OBJECTIVES:
Understand the basic concept of compiler design, and its different phases which will be helpful to construct new tools like LEX, YACC, etc.

COURSE OUTCOMES:
At the end of the course students are able to
CO1: Understand about language processors and its phases and demonstrate about scanning of tokens.
CO2: Perform the syntax analysis by using parsing techniques
CO3: Perform Semantic analysis using attribute grammar
CO4: Compare different memory Management techniques in runtime environment
CO5: Ascertain optimization techniques for intermediate code forms and code generation

UNIT 1
Overview of language processing: – preprocessors – compiler – assembler – Linkers & loaders, difference between compiler and interpreter- structure of a compiler –phases of a compiler. **Lexical Analysis**: - Role of Lexical Analysis – Input Buffering – Specification of Tokens – Recognition of Token – The Lexical Analyzer Generator Lex.

UNIT 2
Syntax Analysis: – Role of a parser, Fuctions of parser, Context Free Grammar. Top Down Parsing: – Recursive Descent Parsing — Non recursive Predictive Parsing- FIRST and FOLLOW – LL(1) Grammar – Error Recovery in Predictive Parsing.

UNIT 3
Bottom up Parsing: – Reductions – Handle Pruning - Shift Reduce Parsing - Introduction to simple LR – Why LR Parsers – Model of an LR Parsers — Construction of SLR Tables.
More powerful LR parsers: - Construction of CLR (1) - LALR Parsing tables.

UNIT 4
Runtime Environment: - Storage organization - Stack allocation – Static allocation - Heap management - Parameter passing mechanisms.
Intermediate code: - DAG - Three address code – Quadruples - Triples - Indirect Triples.

UNIT 5
Common Optimization techniques : - Folding, Copy Propagation, Common Sub expression elimination, Frequency reduction, Strength reduction etc.
Machine dependent code optimization: - Peephole optimization – Register allocation - Instruction scheduling - Inter Procedural Optimization - Garbage collection via reference counting.

Text Books
1.Compilers: Principles, Techniques and Tools: 2nd Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ulman; 2nd Edition, Pearson Education.
2. Modern Compiler Implementation in C- Andrew N. Appel, Cambridge University Press

Reference Books
1. lex &yacc – John R. Levine, Tony Mason, Doug Brown, O‟reilly..
2. Modern Compiler Design- Dick Grune, Henry E. Bal, Cariel T. H. Jacobs, Wiley reamtech.
3. Engineering a Compiler-Cooper & Linda, Elsevier.
4. Compiler Construction, Louden, Thomson. Principles of compiler design, V. Raghavan, 2nd ed, TMH, 2011